



Instruction: Answer All Questions

Time: 90 minutes

Question One

- (a) Explain using different equations why the period, speed and acceleration of an orbiting satellite are not dependent upon the mass of the satellite (2marks)
- (b) Assuming a geostationary satellite wishes to orbit the earth in 24 hours. Where: $M_{\text{earth}} = 5.98 \times 10^{24} \text{ kg}$ and $R_{\text{earth}} = 6.37 \times 10^6 \text{ m}$ determine:
- (i) The radius of the orbit (5marks)
- (ii) The height above the Earth's surface at which it must be located. (5 marks)
- (ii) Enumerate two major advantages of Medium Earth Orbit (MEO) satellites over Geostationary Earth Orbit (GEO) satellites. (2 marks)

Question Two

- (a) To gain a proper understanding of orbital mechanics and launchers in satellite communication, briefly explain the following:
- (i) Orbital perturbation. (2 mark)
- (ii) Launches and launch vehicles. (3 marks)
- (b) Explain the significance of multiple access techniques in satellite communication. Briefly discuss two characteristics of Demand Assigned Multiple Access (DAMA) and provide one example of Very Small Aperture Terminal (VSAT) technologies. (4 marks)
- (c) A geostationary satellite is positioned at 70°W , with an earth station at latitude 45°N and longitude 80°W . Assume Earth's radius, R , is 6371 km, and the mean radius of the geostationary orbit is 41174 km.

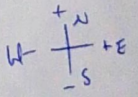
Calculate:

- (i) The antenna look angles. (5 marks)
- (ii) State the conditions for Azimuth angle obtained in (i). (2mark)

$$vel. = \frac{dist}{time} ; \quad time = \frac{dist}{vel.} = \frac{2\pi R}{\sqrt{\frac{GM_e}{R}}}$$

$$\frac{\sqrt{GM_e}}{2\pi} = \sqrt{R^3}$$

$$T = \frac{2\pi R^{3/2}}{(GM_e)^{1/2}}$$





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B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION

COURSE CODE: ICT403 **COURSE TITLE:** Satellite Communication

UNITS: 3

TIME ALLOWED: 180 minutes

Instruction: Answer only Four (4) Questions

Question One

(a) A Low Earth Orbit (LEO) satellite orbits the Earth at a height of 120 km above the surface. Given: M_{earth} , R_{earth} and G are 5.98×10^{24} kg, 6.37×10^6 m and 6.673×10^{-11} Nm²/kg² respectively, determine:

(i) The speed of the satellite, (3marks)

(ii) The orbital period of the satellite. (3marks)

(iii) The Acceleration of the satellite and (3marks)

(iv) Discuss the significance of the acceleration value obtained in ii. (1 mark)

(b) Outline two advantages and two disadvantages of GEO satellite communication. (4 marks)

(c) In simple terms, what sets an active satellite apart from a passive satellite? Support your explanation with two real-life examples for each type. (3 marks)

(d) What exactly is a satellite orbit? Give a short description of three types of satellite orbits, along with what each one is commonly used for. (5 marks)

(e) Using relevant equations, explain why the mass of a satellite does not affect its orbital period, speed, or acceleration. (3 marks)

Question Two

(a) Satellite communication systems rely heavily on efficient sharing of bandwidth. Explain the importance of multiple access techniques in satellite communication and how they influence system capacity and user connectivity. (5 marks)

(b) The Demand Assigned Multiple Access (DAMA) technique plays a vital role in managing dynamic traffic. Discuss the main features of DAMA, highlighting how it enhances flexibility and bandwidth utilization in satellite networks. (5 marks)

(c) Very Small Aperture Terminal (VSAT) systems are widely used in modern satellite networks for both commercial and enterprise services. Describe the core technologies and functionalities integrated into a VSAT system that enable its effective operation. (5 marks)

(d) Communication satellites face a variety of performance-affecting conditions in orbit. Identify and briefly explain five major orbital effects that can impact the quality and reliability of satellite communication systems. (10 marks)



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Question Three

Imagine you are preparing for a panel discussion on satellite systems. Use your knowledge to tackle the following practical and theoretical challenges:

(a) A geostationary satellite takes 12 hours to complete one orbit. Given:
 $M_e = 5.98 \times 10^{24}$ kg, $R_e = 6.37 \times 10^6$ m

Help the team with these:

- (i) Draw a simple, well-labelled sketch showing the Earth and the satellite in orbit (2 marks)
- (ii) Calculate how high above the Earth's surface the satellite is placed (4 marks)
- (iii) Find the radius of the satellite's orbit (3 marks)
- (b) From a technical standpoint, what are three key factors that determine the path (orbit) a satellite will take once launched? (3 marks)
- (c) Medium Earth Orbit (MEO) satellites are gaining wider usage in modern communication systems. State two key advantages they offer over Geostationary Earth Orbit (GEO) satellites. (3 marks)
- (d) Visualize and draw the basic block diagram of a satellite communication system. Give a short explanation of how information flows in both the uplink and downlink directions. (4 marks)
- (e) When assigning frequencies for satellite communications, engineers must be careful. What are three important factors they consider to ensure optimal performance and minimal interference? (6 marks)

Question Four

- (a) As part of a system upgrade, a telecom client has asked you to explain a key concept in TDMA.
 - (i) What is the reference burst, and why is it critical in satellite TDMA systems? (3 marks)
 - (ii) List and describe the major components of a reference burst used in TDMA operations. (6 marks)
- (b) When comparing access techniques in satellite systems, CDMA is often favoured over FDMA in certain scenarios. Highlight three compelling advantages that CDMA offers over FDMA in satellite communication. (3 marks)
- (c) Imagine you're designing a secure satellite link for military or mobile use. Differentiate between Frequency Hopping CDMA (FH-CDMA) and Time Hopping CDMA (TH-CDMA). For each, give a real-life example and where it's applied. Which one would you choose, and why? (5 marks)
- (d) Satellite networks often rely on random access protocols to manage data transmission. Compare Pure ALOHA and Slotted ALOHA, clearly explaining how each works, their efficiencies, and which is better suited for satellite communication — with reasons. (8 marks)



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QUESTION Five

(a) A Medium Earth Orbit (MEO) satellite is positioned at longitude 70°W , while an Earth station is located at latitude 45°N and longitude 80°W .

Assume the Earth's radius is 6,371 km, and the satellite's orbital radius is 26,571 km.
Using these details:

(i) Calculate the **slant range (distance)** between the Earth station and the satellite. (3 marks)

(b) With reference to the same scenario:

(i) Determine the azimuth angle for proper antenna alignment. (4 marks)

(ii) Determine the elevation angle from the Earth station to the satellite. (5 marks)

(iii) State the conditions for accurately determining azimuth angles in satellite communication, and explain how these conditions affect the elevation angle in (ii). (5 marks)

(c) In the context of satellite-earth station alignment, explain the importance of look angles (azimuth and elevation) to the performance of an Earth station antenna. (3 marks)

(d) Multibeam satellites are increasingly used for improved coverage and capacity.
List three key advantages and two limitations of using multibeam satellite systems. (5 marks)

QUESTION Six

(a) To build a strong foundation in orbital mechanics and satellite launch systems, briefly explain the following key concepts:

(i) Satellite launches and launch vehicles (6 marks)

(ii) Orbital determination (3 marks)

(iii) Orbital perturbation (3 marks)

(iv) Look angle determination (2 marks)

(b) Highlight three emerging trends in satellite communication and three key advantages that support its continued growth. (8 marks)

(c) State Kepler's laws of planetary motion and explain their relevance to the motion of artificial satellites. (3 marks)